# Ro02: Probability Primer and Recursive Bayes Filter

## Probabilistic Approach

- Use of probability theory to explicitly represent uncertainty (robot motion and observations)
  - Gaussian/Normal Distribution
- Causes:
  - Key tool for robust state estimation and robotics
  - model uncertainties

Discrete Random Variables : probability for each countable number of values with probability mass function
<->
Continuous Random Variable: Probability Density Function for continuous number range

- Probability sums up to One!

## Joint and Conditional Probability

- $P(X = x \text{ and } Y = y) = P(x, y)$
- If X and Y are **independent** then

$$P(x, y) = P(x)\,P(y)$$

- $P(x \mid y)$ is the probability of ***X* given *Y***

$$P(x, y) = P(x \mid y)\,P(y)$$

$$P(x \mid y) = P(x, y)/P(y)$$

- If X and Y are **independent** then

$$P(x \mid y) = P(x)$$

### Law of Total Probability

- Discrete case:
  - $P(x) = \sum_y P(x|y)P(y)$
  - Marginalisation : $P(x) = \sum_y P(x, y)$

- Continuous case:
  - $p(x) = \int p(x|y)p(y)dx$
  - Marginalisation : $p(x) = \int p(x, y)dy$

# Example

| | $M$ | $\neg M$ | $\sum$ |
|---|---|---|---|
| $S$ | 20 | 20 | 40 |
| $\neg S$ | 35 | 45 | 80 |
| $\sum$ | 55 | 65 | 120 |

$$P(M, S) = \frac{20}{120} \qquad P(M, \neg S) = \frac{35}{120} \qquad \cdots$$

$$P(M) = \frac{55}{120} \qquad P(\neg M) = \frac{65}{120} \qquad P(S) = \frac{40}{120} \qquad \cdots$$

$$P(M \mid S) = \frac{P(M, S)}{P(S)} = \frac{20/120}{40/120} = \frac{20}{40} \qquad \cdots$$

## Bayes Rule with Background Knowledge

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$
$$\rightarrow P(x|y, z) = \frac{P(y|x, z)P(x|z)}{P(y|z)}$$

# Conditional Independence

Two variables are conditional independent if

$$P(x, y \mid z) = P(x \mid z)\, P(y \mid z)$$

- Equivalent to $P(x \mid z) = P(x \mid z, y)$
  and $P(y \mid z) = P(y \mid z, x)$

- But this does **not imply**

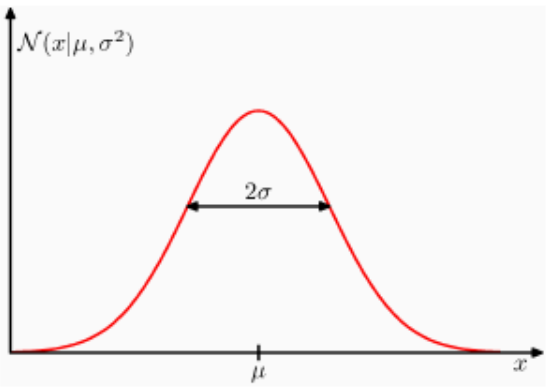
$$P(x, y) = P(x)\, P(y)$$

(independence/marginal independence)

Normal Distribution

- Univariate: 1-dimensional
- Unimodal: one mode

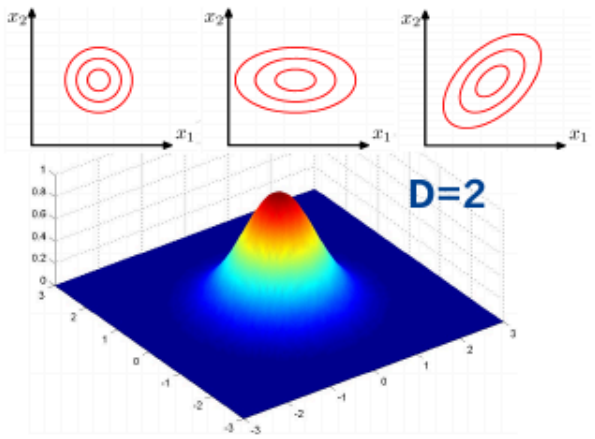
$$\mathcal{N}\left(x|\mu, \sigma^2\right) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}$$

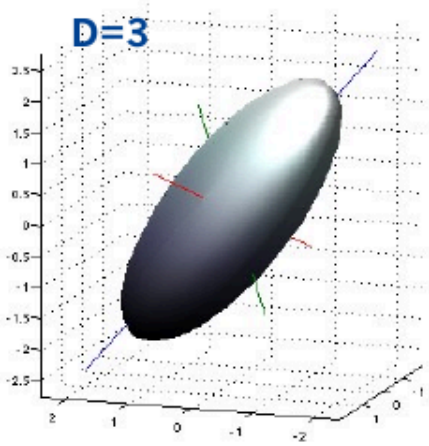


# Multivariate Normal Distribution

- Multivariate: multi-dimensional
- Unimodal: one mode

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}$$

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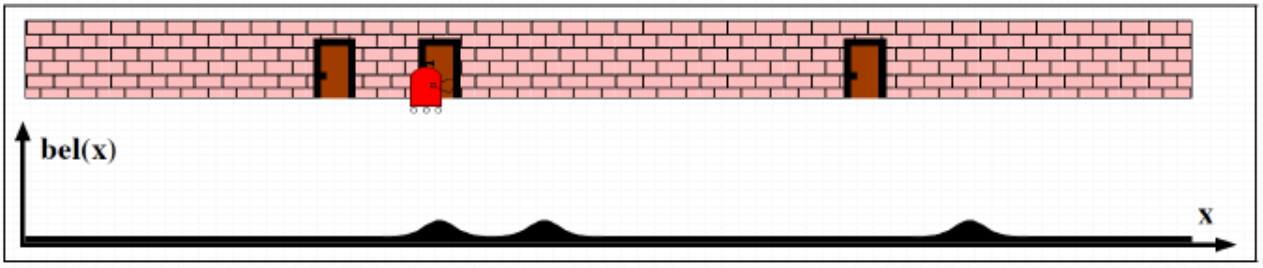
## Recursive State Estimation

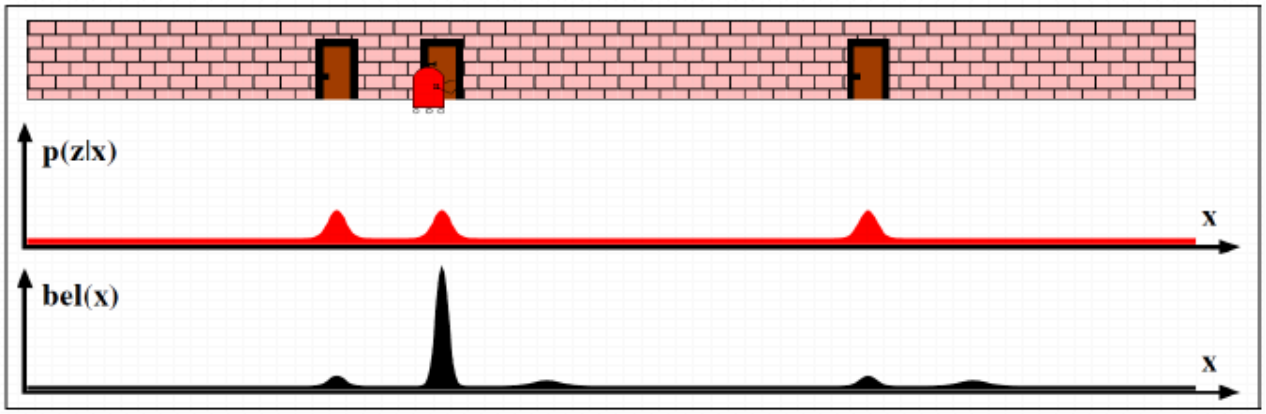
Estimate $x$ of a system given observations $z$ and controls $u$

$$p(x|z, u)$$

...

$$p(x_t|z_{1:t}, u_{1:t})$$




## Complete Derivation of Recursive Bayes Filter

$$
\begin{aligned}
bel(x_t) &= p(x_t|z_{1:t}, u_{1:t}) && \text{definition of the belief} \\
&= \eta \cdot p(z_t|x_t, z_{1:t-1}, u_{1:t}) p(x_t|z_{1:t-1}, u_{1:t}) && \text{Bayes' rule} \\
&= \eta \cdot p(z_t|x_t) p(x_t|z_{1:t-1}, u_{1:t}) && \text{Markov assumption} \\
&= \eta \cdot p(z_t|x_t) \int p(x_t|x_{t-1}, z_{1:t-1}, u_{1:t}) p(x_{t-1}|z_{1:t-1}, u_{1:t}) dx_{t-1} && \text{Law of total probability} \\
&= \eta \cdot p(z_t|x_t) \int p(x_t|x_{t-1}, u_t) p(x_{t-1}|z_{1:t-1}, u_{1:t}) dx_{t-1} && \text{Markov assumption} \\
&= \eta \cdot p(z_t|x_t) \int p(x_t|x_{t-1}, u_t) p(x_{t-1}|z_{1:t-1}, u_{1:t-1}) dx_{t-1} && \text{Independence assumption} \\
&= \eta \cdot p(z_t|x_t) \int p(x_t|x_{t-1}, u_t) \cdot bel(x_{t-1})) dx_{t-1} && \text{Markov assumption}
\end{aligned}
$$

### Prediction and Correction Step

- Bayes filter written as 2-Step process
  - $bel(x_t) = \eta \cdot p(z_t|x_t) \qquad \int p(x_t|x_{t-1}, u_t) \cdot bel(x_{t-1})) dx_{t-1}$
- **Prediction step**
  - $\overline{bel}(x_t) = \int \underbrace{p(x_t|x_{t-1}, u_t)}_{\text{motion model}} \cdot bel(x_{t-1})) dx_{t-1}$
- **Correction step**
  - $bel(x_t) = \eta \cdot \underbrace{p(z_t|x_t)}_{\text{observation model}} \quad \overline{bel}(x_t)$
  - Observation/ measurement/ sensor model

### Different Realisations

- Bayes filter is a **framwork** for recursive state estimation
- Different realisations
- Different properties
  - Linear vs. non-linear models for motion and observation models
  - Gaussian distributions only?
  - Parametric vs. non-parametric filters
  - ...

### Popular Filters

- **Kalman filter & EKF**
  - Gaussians
  - Linear or linearised models
- **Particle filter**
  - Non-parametric
  - Arbitrary models (sampling required)

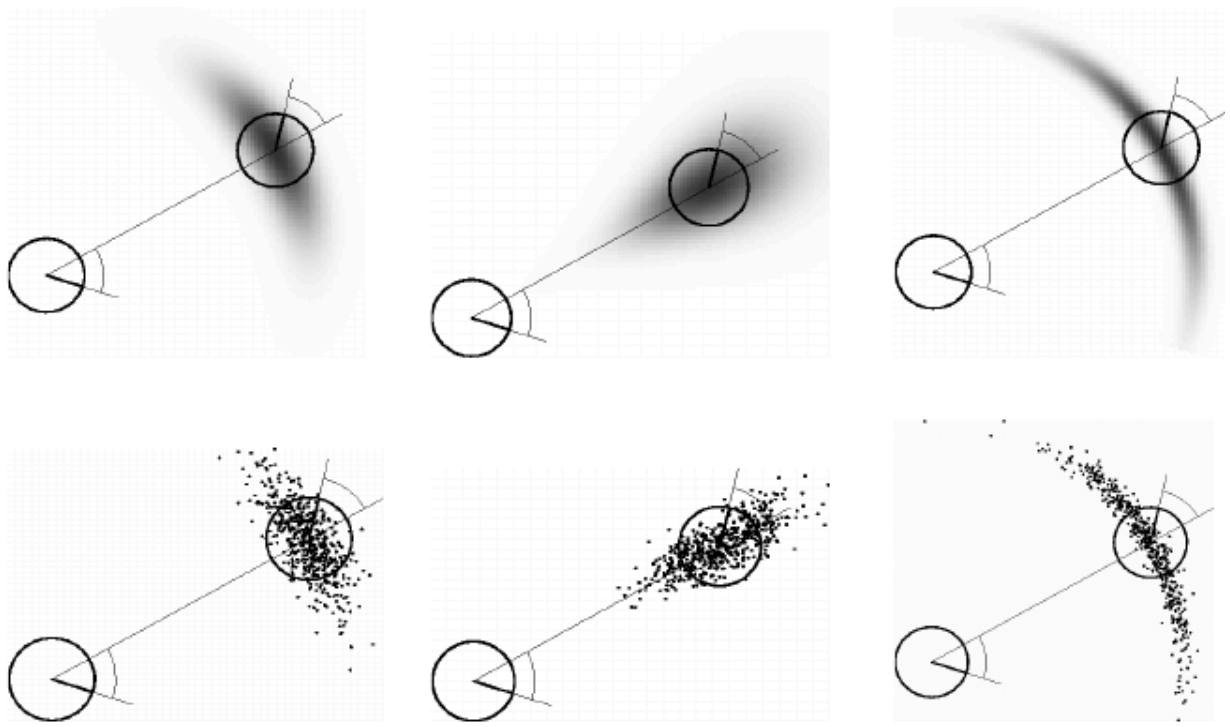
## Motion Model

$$\overline{bel}(x_t) = \int \underbrace{p(x_t|x_{t-1}, u_t)}_{\text{motion model}} \cdot bel(x_{t-1}))dx_{t-1}$$

#### Basic Motion Models

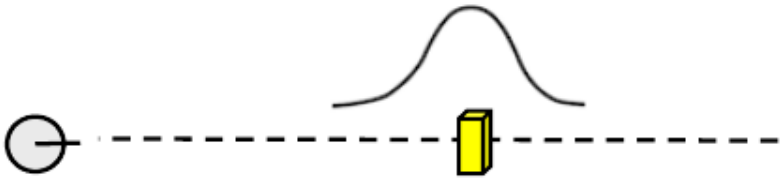
- Motion is inherently uncertain
- How to model uncertainty?

### Example: Odometry-Based Motion



# Observation Model

$$bel(x_t) = \eta \cdot \underbrace{p(z_t|x_t)}_{\text{observation model}} \overline{bel}(x_t)$$

- Via Range sensors
- Example: Simple Observation Model with Gaussian Noise
  - Range sensors estimating the distance to the closest obstacle
  - Gaussian noise in the range reading
  - 

## Summary

- Probabilities occur for most of the problems addressed
- Bayes filter is framework for state estimation
- $\exists$ different realisations (EKF, particle filter,...)
- Motion and observation models are central models in Bayes filter
  - Need to be specified !